

NORTHERN UNIVERSITY OF BUSINESS & TECHNOLOGY KHULNA

Department of Computer Science and Engineering
Project Documentation

Course Title: Mobile Computing LAB

Course Code: CSE 4204

Project Name: Medical Report Analyzer **Document:**
Frontend Progress Report

Submitted To:

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Northern University of Business and Technology Khulna

Submitted By:

Team: 02

Section: 8D

Submission Date: 13 August , 2026

AI Feature Interface

Fig : Dashboard

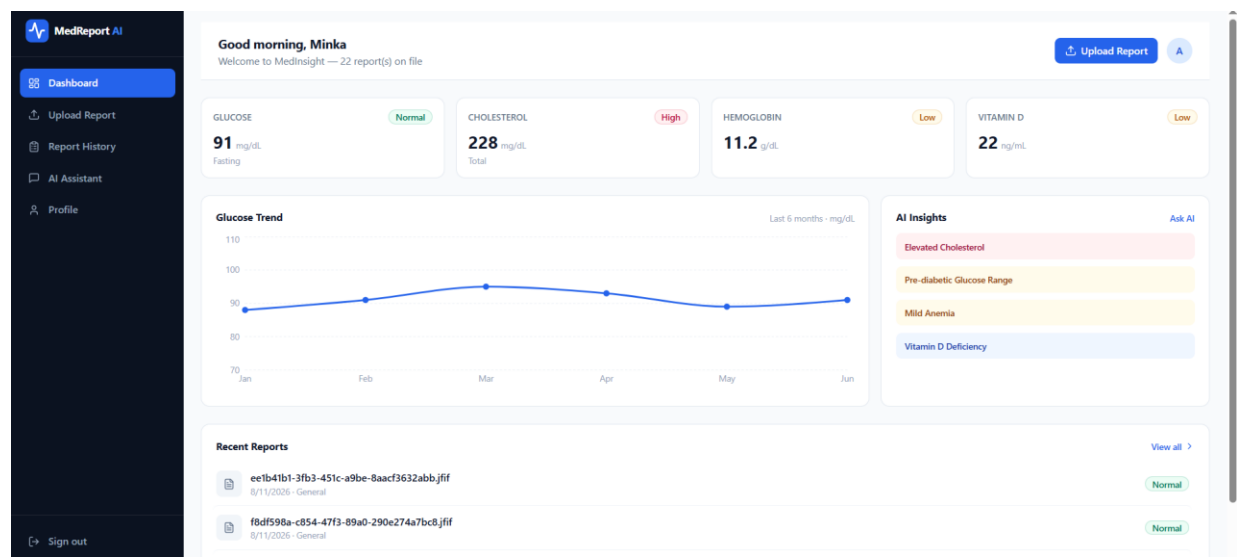


Fig : AI Assistant

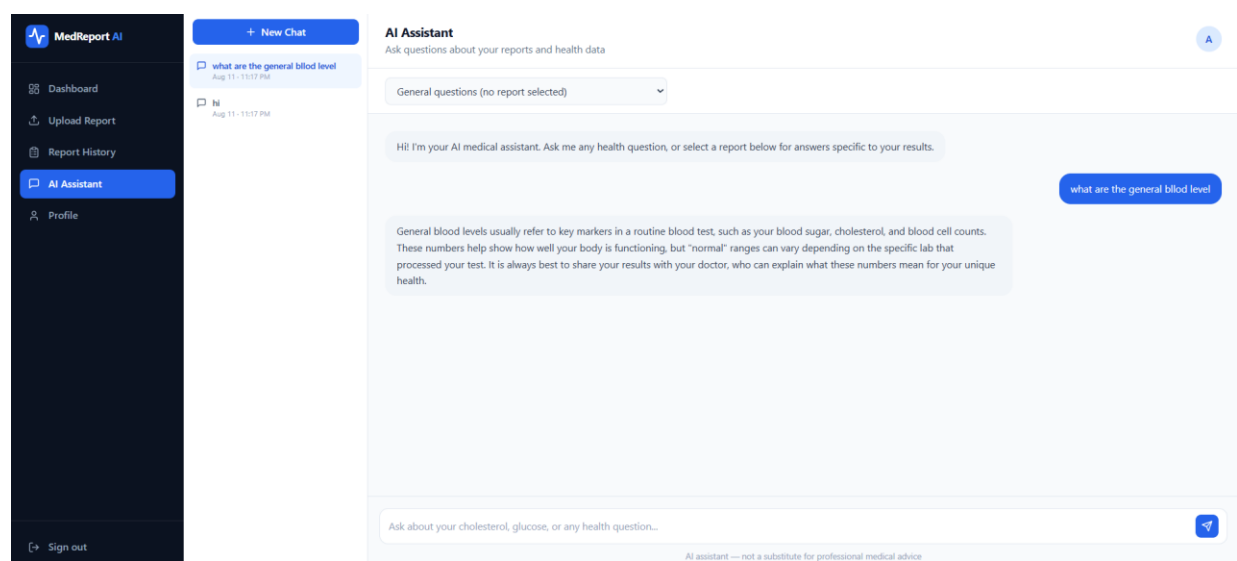


Fig : Result pages

MedReport AI

Dashboard

Upload Report

Report History

AI Assistant

Profile

Sign out

Back

ExportAll Reports

ee1b41b1-3fb3-451c-a9be-8aacf3632abb.jfif

8/11/2026 · General

AI HEALTH SUMMARY · PLAIN LANGUAGE

The patient's lab results indicate excellent overall cardiovascular and liver health, with cholesterol, triglycerides, and liver enzymes all well within normal limits. The blood sugar levels after the glucose tolerance test are also normal, though the initial blood sugar level of 3.1 mmol/L is slightly lower than the standard fasting range. It is recommended to discuss this low initial reading with a doctor to rule out mild hypoglycemia.

7 Normal0 High1 Low

8TOTAL MARKERS

Detected Values

MARKER	VALUE	RANGE	STATUS
Blood Sugar (Initial)	3.1 mmol/L	—	Low
Blood Sugar (1 hr after 75g glucose)	4.8 mmol/L	—	Normal
Blood Sugar (2 hr after 75g glucose)	5.2 mmol/L	—	Normal
SGPT (ALT)	32 U/L	—	Normal
Serum Cholesterol	163 mg/dl	—	Normal
Serum Triglycerides	117 mg/dl	—	Normal
Serum HDL (Good Cholesterol)	50 mg/dl	—	Normal
Serum LDL (Bad Cholesterol)	89 mg/dl	—	Normal

Risk Warnings

Blood Sugar (Initial)

The initial blood sugar level of 3.1 mmol/L is below the typical healthy fasting range (usually 3.9 to 5.6 mmol/L), which may suggest mild hypoglycemia.

Recommendations

01. Schedule a follow-up with your doctor within 4 weeks

02. Reduce saturated fat and refined carbohydrate intake

03. Aim for 30 min of aerobic exercise, 5+ days per week

04. Eat iron-rich foods: spinach, lean red meat, legumes

05. Take a Vitamin D3 supplement (1,000-2,000 IU/day)

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ExportAll Reports

4f7c30cb-6f38-47e1-8964-60d290f98f82.jfif

8/11/2026 · CBC

AI HEALTH SUMMARY · PLAIN LANGUAGE

This hormone report for 25-year-old Minka indicates a hormone level (likely TSH) of 0.38, which is slightly below the standard reference range. This borderline low result suggests the need for a comprehensive thyroid evaluation to rule out mild thyroid overactivity.

0 Normal0 High1 Low

1TOTAL MARKERS

Detected Values

MARKER	VALUE	RANGE	STATUS
Hormone Level (Likely TSH)	0.38	—	Low

Risk Warnings

Hormone Level (Likely TSH)

The value of 0.38 is slightly below the typical lower limit of normal (usually 0.4 uIU/mL), which can sometimes point to an overactive thyroid.

Recommendations

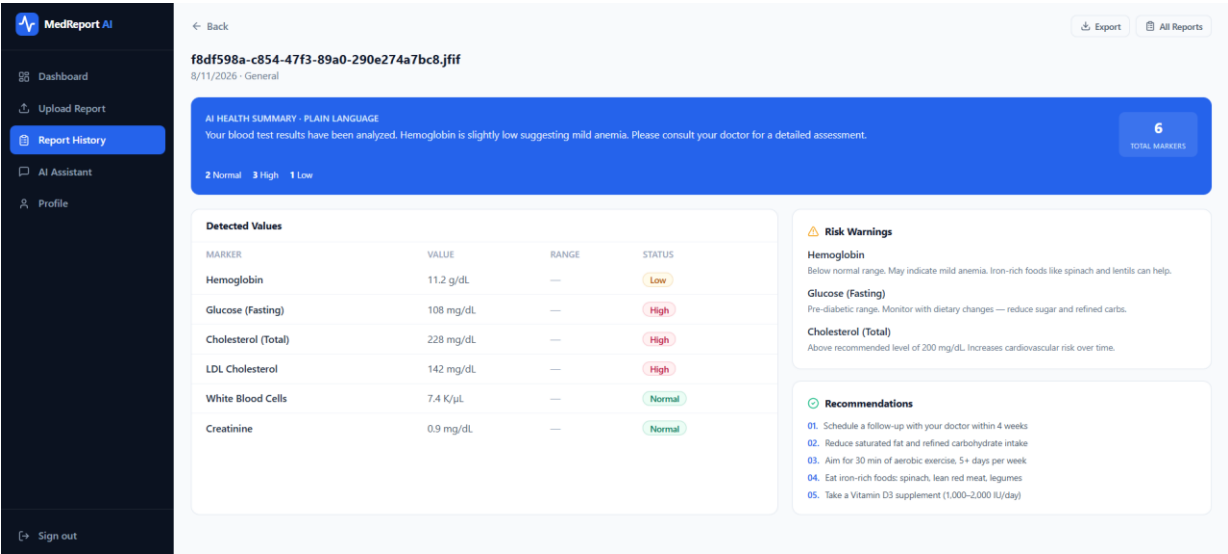
01. Schedule a follow-up with your doctor within 4 weeks

02. Reduce saturated fat and refined carbohydrate intake

03. Aim for 30 min of aerobic exercise, 5+ days per week

04. Eat iron-rich foods: spinach, lean red meat, legumes

05. Take a Vitamin D3 supplement (1,000-2,000 IU/day)



AI Interaction

Fig : General Health Questions

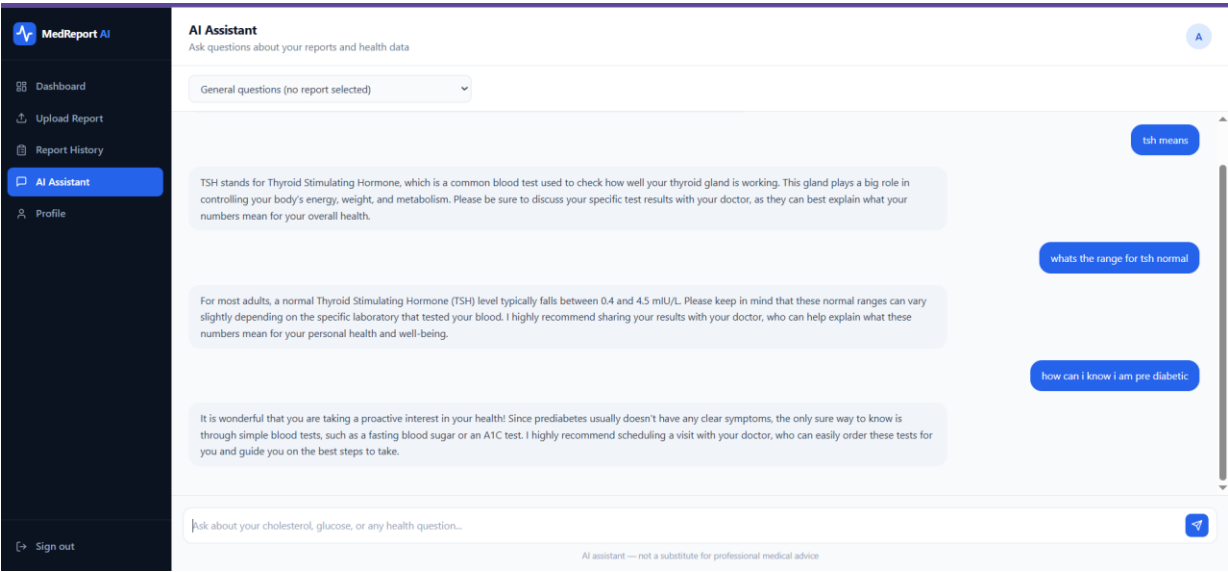
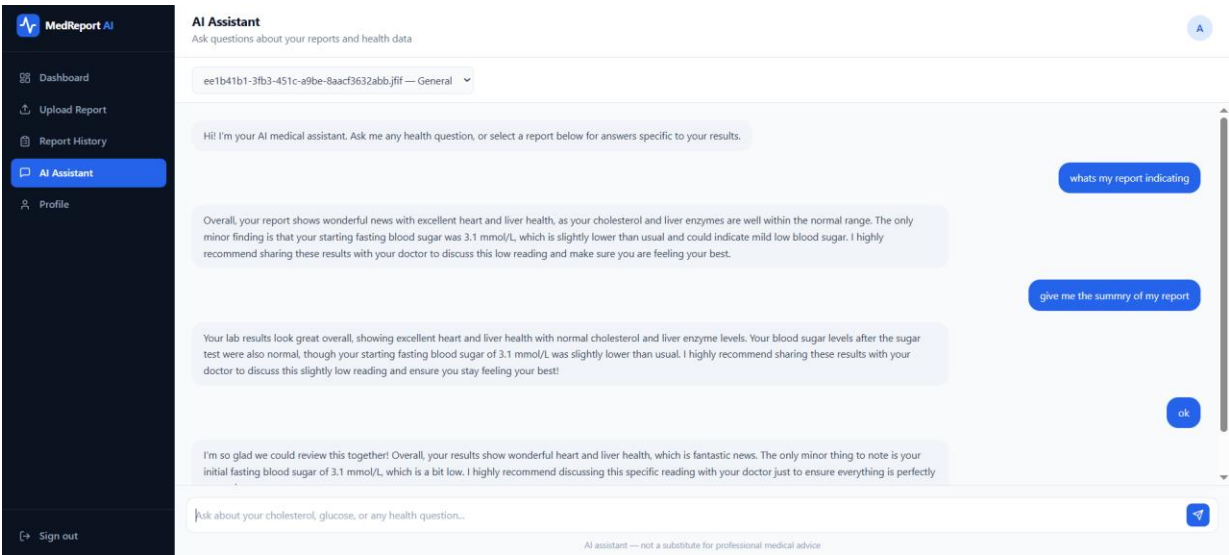


Fig : Report-Context Selected Questions



AI Responses

Fig : AI Summary

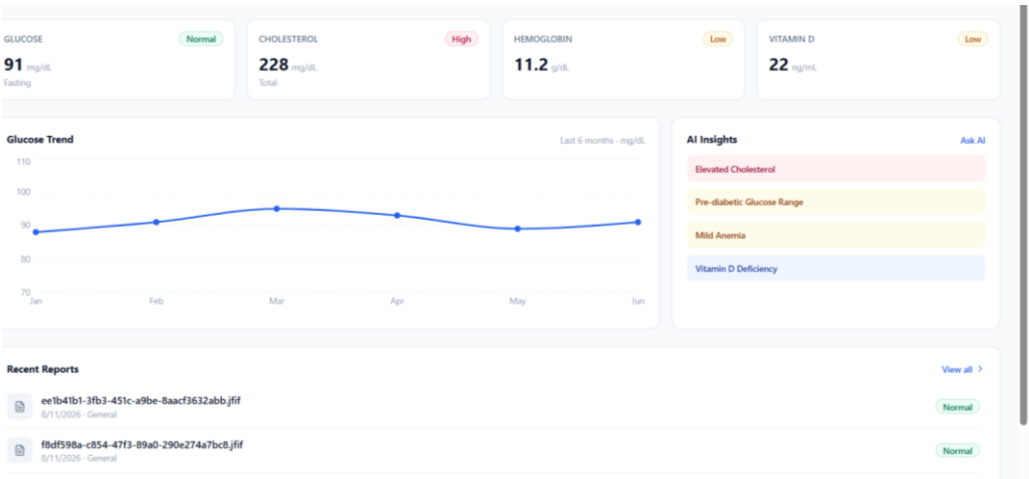


Fig : AI Risk Warning



Risk Warnings

Blood Sugar (Initial)

The initial blood sugar level of 3.1 mmol/L is below the typical healthy fasting range (usually 3.9 to 5.6 mmol/L), which may suggest mild hypoglycemia.

Fig : Export AI Response

← Back

4f7c30cb-6f38-47e1-8964-60d290f98f82.jfif

8/11/2026 · General

AI HEALTH SUMMARY · PLAIN LANGUAGE

This is a hormone blood report for 25-year-old Minka. The document contains highly obscured text, but indicates a hormone evaluation with a TSH (Thyroid Stimulating Hormone) level of approximately 0.38 uIU/mL, which is borderline low. A physician should review the complete physical report to confirm these findings.

0 Normal

0 High

1 Low

Detected Values

MARKER	VALUE
TSH (Thyroid Stimulating Hormone)	0.38 uIU/mL

Export

All Reports

1

TOTAL MARKERS

03. Aim for 30 min of aerobic exercise, 5+ days per week

04. Eat iron-rich foods: spinach, lean red meat, legumes

05. Take a Vitamin D3 supplement (1,000–2,000 IU/day)

4f7c30cb-6f38-47e1-8964-60d290f98f82-summary (2).txt

File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ? + - X

1 MedInsight AI — Report Export

2 =====

3 Report: 4f7c30cb-6f38-47e1-8964-60d290f98f82.jfif

4 Date: 8/11/2026

5 Type: General

6

7 AI HEALTH SUMMARY

8 This is a hormone blood report for 25-year-old Minka. The document contains highly obscured text, but indicates a hormone evaluation with a TSH (Thyroid Stimulating Hormone) level of approximately 0.38 uIU/mL, which is borderline low. A physician should review the complete physical report to confirm these findings.

9

10 DETECTED VALUES (0 Normal, 0 High, 1 Low)

11 - TSH (Thyroid Stimulating Hormone): 0.38 uIU/mL [Low]

12

13 RISK WARNINGS

14 - TSH (Thyroid Stimulating Hormone): The value of 0.38 uIU/mL is slightly below the typical normal lower limit of 0.4 uIU/mL, which can sometimes point to an overactive thyroid (hyperthyroidism) and

length:1,202 line Ln:1 Col:1 Pos:1

Unix (LF) UTF-8 INS

Prompts Examples

Fig : AI Prompts Shown

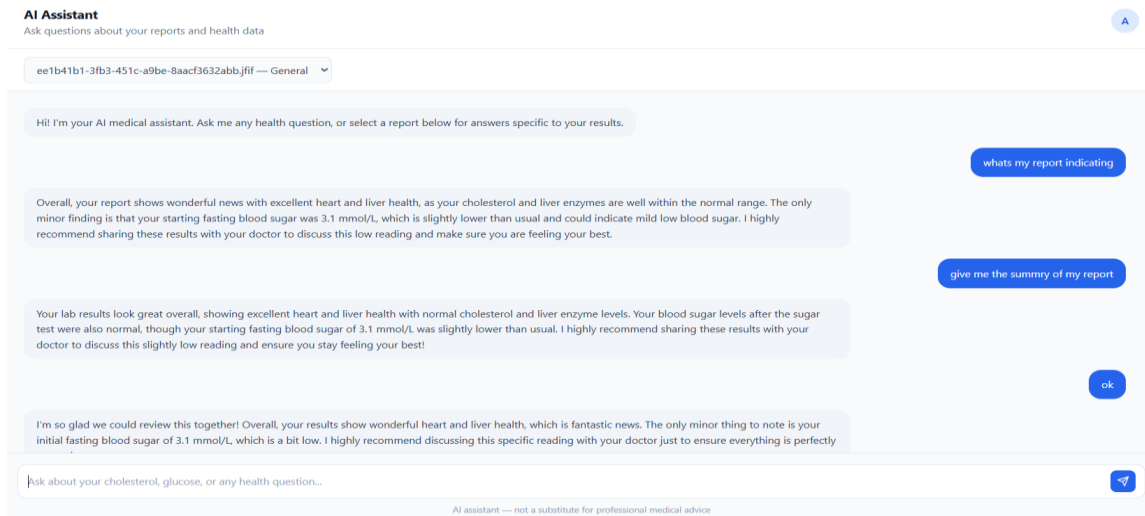
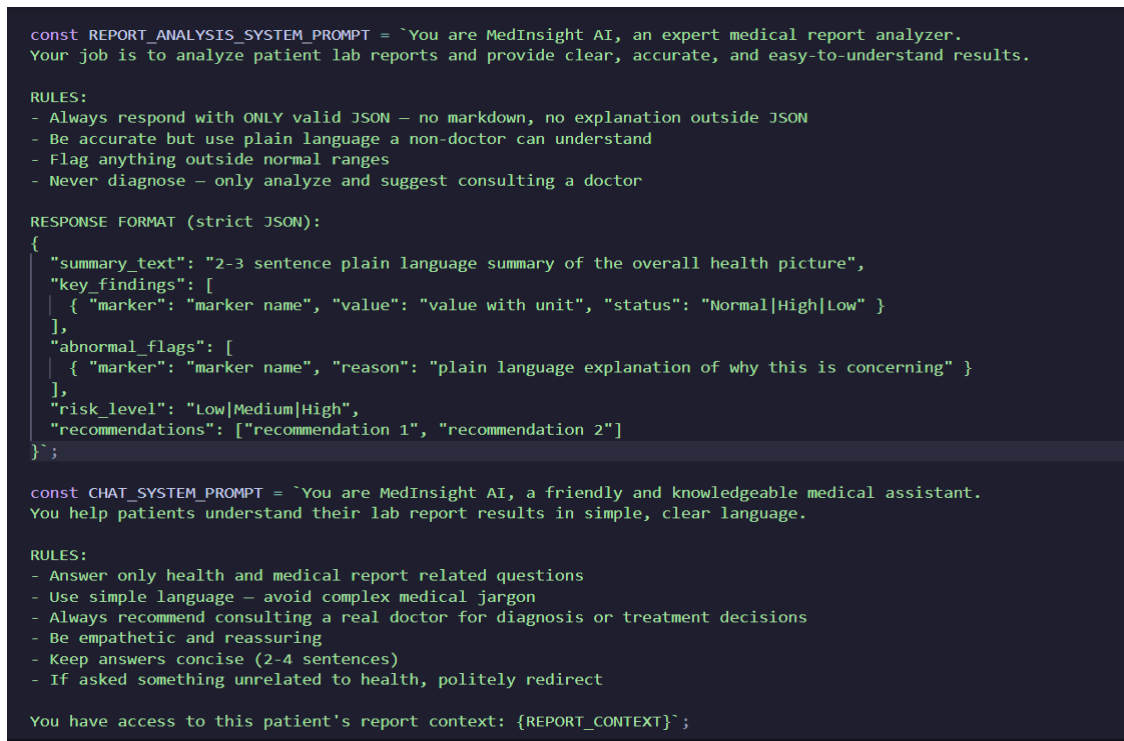


Fig : Prompt Engineering



Backend Integration

Fig: Terminal/console log -during a live request

```
PS C:\Users\User\Desktop\medinsight-backend> cd C:\Users\User\Desktop\medinsight-backend\medinsight-backend
PS C:\Users\User\Desktop\medinsight-backend\medinsight-backend> npm run dev

> medinsight-backend@1.0.0 dev
> nodemon server.js

[nodemon] 3.1.14
[nodemon] to restart at any time, enter `rs`
[nodemon] watching path(s): *.*
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting `node server.js`
MedInsight API running on port 5000 [development]

PS C:\Users\User\Desktop\medinsight-backend\medinsight-backend> npm run dev
MedInsight API running on port 5000 [development]
[DEBUG] Gemini model in use: gemini-3.5-flash
[DEBUG] Raw Gemini response (analyzeReport): {
  "summary_text": "The lab report indicates a hormone analysis for 25-year-old Minka. The thyroid-stimulating hormone (TSH) level appears to be slightly below the standard reference range. Further evaluation is recommended to assess overall thyroid function.",
  "key_findings": [
    {
      "marker": "TSH (Thyroid Stimulating Hormone)",
      "value": "0.38 uIU/mL",
      "status": "Low"
    }
  ],
  "abnormal_flags": [
    {
      "marker": "TSH (Thyroid Stimulating Hormone)",
      "reason": "The TSH level is slightly below the typical normal range, which can sometimes suggest an overactive thyroid gland (hyperthyroidism) and warrants further medical review."
    }
  ],
  "risk_level": "Medium",
  "recommendations": [
    "Consult your referring physician, Dr. Debasish Kumar Ghosh, to discuss these findings.",
    "Consider undergoing a complete thyroid panel, including Free T3 and Free T4, for a comprehensive assessment.",
    "Monitor for any physical symptoms such as unexplained weight changes, rapid heart rate, anxiety, or temperature sensitivity."
  ]
}
```


Fig: GitHub Repository- AI Integration Branch

CSE-4204-8D-T02Public

Watch0

AI-Integration-_Farjana-Yasmin-Minka

4 Branches0 Tags

Go to fileT

Add file

<> Code

This branch is 2 commits ahead of and 63 commits behind main.

Contribute

minka

Move AI integration backend into backend directory

b4afa7d · 31 minutes ago

2 Commits

backend	Move AI integration backend into backend directory	31 minutes ago
.env.example	AI Integration implementation	1 hour ago
.env.temp	AI Integration implementation	1 hour ago
.gitignore	AI Integration implementation	1 hour ago
README.md	AI Integration implementation	1 hour ago
my-current-changes.patch	AI Integration implementation	1 hour ago
package-lock.json	AI Integration implementation	1 hour ago
package.json	AI Integration implementation	1 hour ago
server.js	AI Integration implementation	1 hour ago

README

CSE-4204-8D-T02 / backend /

Add file...

minka

Move AI integration backend into backend directory

b4afa7d · 31 minutes ago

History

This branch is 2 commits ahead of and 63 commits behind main.

Contribute

Name	Last commit message	Last commit date
..		
config	Move AI integration backend into backend directory	31 minutes ago
controllers	Move AI integration backend into backend directory	31 minutes ago
middleware	Move AI integration backend into backend directory	31 minutes ago
models	Move AI integration backend into backend directory	31 minutes ago
routes	Move AI integration backend into backend directory	31 minutes ago
tests	Move AI integration backend into backend directory	31 minutes ago
utils	Move AI integration backend into backend directory	31 minutes ago